12TH BIOLOGY · CHAPTER 17 · FULL CHAPTER NOTES

Human Reproductive Systems

Topic-wise short and long questions with answers from the complete chapter

17.1 REPRODUCTIVE SYSTEMS

Q.1 [Short] What are the functions of the reproductive systems?

The functions of reproductive systems include the production of reproductive cells (gametes) and the preparation of gametes for fertilization. In addition, the male reproductive system delivers the gametes to the female reproductive system. The female reproductive system nourishes the fertilized egg cell and provides an environment for its development into a baby.

Q.2 [Short] In which two aspects is the reproductive system different from other organ systems?

1. The reproductive system is quite different in male and female, while the other organ systems are almost the same.
2. The reproductive system does not become functional until puberty. All other body systems are functional since or shortly after birth.

17.1.1 MALE REPRODUCTIVE SYSTEM

Q.3 [Long] Describe the structure of the human male reproductive system and the function of each part.

1. Gonads (testes)

Two testes are located outside the abdomen in a skin pouch called the **scrotum**. Each testis is oval-shaped, about 3–5 cm long and 2–3 cm wide. A testis is composed of highly convoluted **seminiferous tubules**. The wall of the seminiferous tubules, i.e., the germinal epithelium, contains two types of cells: **germline cells**, which make sperms (spermatozoa) by meiosis, and **Sertoli cells**, which nourish the developing sperms.

2. Accessory ducts

From the seminiferous tubules, the spermatozoa enter a network of tubules called the **rete testis**. From here they enter a long, coiled tube called the **epididymis**. Here the spermatozoa are still non-motile. They remain in the epididymis for at least 18 hours. Then they become motile and enter another long tube, the **vas deferens**, which passes into the abdominal cavity via the inguinal canal.

The vas deferens from each testis joins with a duct from a seminal vesicle. After joining, the tube is called the **ejaculatory duct**. The two ejaculatory ducts pass through the prostate gland. Here they open into the **urethra**, which comes from the urinary bladder. The urethra opens out of the body through the penis.

3. Accessory glands

These are the glands that add fluids to sperms to form semen.

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1. **Seminal vesicles:** There are two seminal vesicles. Each releases a fluid rich in fructose (sugar) into the vas deferens through a duct. This fluid provides energy for the sperm.
2. **Prostate gland:** It lies just below the urinary bladder. It is soft and spongy and produces about 60% of the fluid of the semen.
3. **Bulbourethral glands:** There are two small bulbourethral glands. They secrete a clear fluid that lubricates the urethra for sperm passage and provides a protective neutral environment for sperms to survive.

4. Copulatory organ (penis)

The male copulatory organ is called the penis. It contains the urethra, erectile tissue, an artery and two veins. It delivers the semen to the female reproductive system.

Q.4 [Short] Why are the testes located outside the abdomen in the scrotum?

The scrotum maintains the testes at around 34°C, which is slightly lower than the core body temperature (37°C). This lower temperature is required for the development of sperms. During development, the testes are formed within the abdominal cavity and descend into the scrotum shortly before birth.

Q.5 [Short] Differentiate between germline cells and Sertoli cells.

Both are present in the germinal epithelium (wall) of the seminiferous tubules. Germline cells make sperms by meiosis. Sertoli cells nourish the developing sperms.

Q.6 [Short] What happens to sperms in the epididymis?

Spermatozoa entering the epididymis are non-motile. They remain there for at least 18 hours, become motile, and then enter the vas deferens.

Q.7 [Short] What is semen?

Semen is a mixture of sperms and the fluids from the seminal vesicles and the prostate gland. The prostate gland produces about 60% of the fluid of the semen.

Q.8 [Short] Trace the path of a sperm from the seminiferous tubules to the outside.

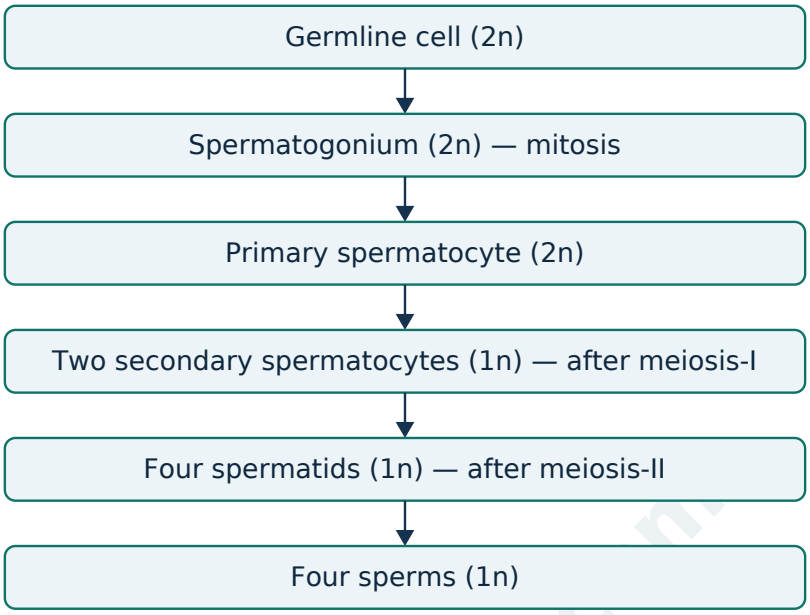
Seminiferous tubules → Rete testis → Epididymis → Vas deferens → Ejaculatory duct → Urethra (through the penis) → Outside.

Q.9 [Long] What is spermatogenesis? Describe its steps.

Spermatogenesis is the process by which sperms are formed inside the seminiferous tubules of the testes.

1. In the walls of the seminiferous tubules, germline cells (2n) divide by mitosis. Some of the new cells become **spermatogonia** (2n), while others remain as germline cells for future use.
2. The spermatogonia continue to divide and form **primary spermatocytes** (2n), which move toward the lumen of the tubules.
3. Each primary spermatocyte divides by meiosis-I to form two **secondary spermatocytes** (1n).

4. These cells quickly undergo meiosis-II to produce **spermatids** (1n). In this way, one spermatogonium (2n) produces four spermatids (1n).
5. Each spermatid matures into an elongated **sperm**.



Spermatogenesis

Q.10 [Short] Describe the structure of a sperm.

A sperm consists of three parts: a head, a midpiece and a tail.

- **Head:** The genetic material is present within the head. About two-thirds of the head is surrounded by a thick cap called the **acrosome**. It is formed mainly of Golgi body and contains enzymes that later help the sperm to fertilize the ovum.
- **Midpiece:** It contains mitochondria and a centriole.
- **Tail:** It is formed of microtubules (axial filament), which arise from the centriole of the midpiece.

The mature spermatozoa in the lumen of the seminiferous tubules are non-motile. They enter the epididymis with the help of the fluid secreted by Sertoli cells. In the epididymis they gain motility.

Q.11 [Short] What is the acrosome? What is its function?

The acrosome is a thick cap that surrounds about two-thirds of the head of a sperm. It is formed mainly of Golgi body. It contains enzymes that help the sperm to fertilize the ovum.

Q.12 [Short] Why does the midpiece of a sperm contain mitochondria?

The tail of the sperm needs energy to move. The mitochondria of the midpiece supply this energy (ATP), so that the sperm can swim to the ovum.

Q.13 [Long] Explain the hormonal control of the male reproductive system.

Spermatogenesis is controlled by the hormones of the hypothalamus and the pituitary gland.

- **GnRH:** The hypothalamus releases gonadotropin-releasing hormone (GnRH). It stimulates the anterior pituitary to release FSH and LH.
- **FSH:** Follicle-stimulating hormone stimulates the seminiferous tubules for spermatogenesis.
- **LH:** Luteinizing hormone stimulates the Leydig cells (located in the interstitial tissue between the seminiferous tubules) to secrete testosterone.
- **Testosterone:** It stimulates the development of the germinal epithelium and maintains the male secondary sexual characteristics.
- **Inhibin:** FSH causes the Sertoli cells to release a peptide hormone called inhibin, which inhibits further FSH secretion (negative feedback).
- Similarly, the release of testosterone inhibits the release of LH and GnRH (negative feedback).

Q.14 [Short] What is inhibin?

Inhibin is a peptide hormone released by the Sertoli cells under the influence of FSH. It inhibits further FSH secretion from the anterior pituitary.

Q.15 [Short] How is the testosterone level controlled by negative feedback?

LH stimulates the Leydig cells to secrete testosterone. The released testosterone inhibits the release of LH from the anterior pituitary and GnRH from the hypothalamus. So less testosterone is then secreted.

17.1.2 FEMALE REPRODUCTIVE SYSTEM

Q.16 [Short] Name the parts of the female reproductive system. Which are accessory organs?

It consists of gonads (ovaries), oviducts, uterus and vagina. The oviducts, uterus and vagina are called the accessory organs of the female reproductive system.

Q.17 [Short] Describe the structure and function of the ovaries.

Ovaries produce egg cells (ova or oocytes) and release hormones. There are two ovaries, each present on the side of the uterus. An oval-shaped ovary is 3–5 cm long and 2–3 cm wide. It contains many tiny sac-like structures called **ovarian follicles**, which are the functional units of the ovary. Follicles nurture immature eggs. In an adult female, one of the ripening follicles ejects its egg cell each month. This event is called **ovulation**. After ovulation, the ruptured follicle is transformed into a glandular structure called the **corpus luteum**.

Q.18 [Short] Write a note on the oviducts.

The oviducts (fallopian tubes or uterine tubes) are two tubes leading from the ovaries into the uterus. When an ovum matures, the ovarian follicle and ovary wall rupture to release it. The ovum enters an oviduct and travels toward the uterus. This movement is helped by the cilia on the inner lining of the oviduct. Fertilization of the ovum also occurs in the oviduct. The resulting zygote undergoes several mitotic divisions while still in the oviduct.

Q.19 [Long] Describe the structure of the uterus.

The uterus is a hollow and muscular organ into which the two oviducts enter. It consists of three parts: the **fundus** (uppermost rounded portion above the openings of the oviducts), the **body** (major portion) and the **cervix** (lower narrow part). The opening of the cervix leads to the vagina.

The wall of the uterus has three layers. The **perimetrium** is the outermost thin layer. The **myometrium** is the middle thick layer of smooth muscles. The **endometrium** is the inner spongy lining. It has a basal layer and a functional layer. The basal layer has stem cells which regenerate the functional layer. The functional layer thickens and is shed during the menstrual cycle. During pregnancy, it thickens further to implant the embryo and forms the **placenta**, which supplies oxygen and nutrition to the embryo and foetus.

Q.20 [Short] Differentiate between the basal layer and the functional layer of the endometrium.

The basal layer has stem cells which regenerate the functional layer. The functional layer thickens and is shed during the menstrual cycle; during pregnancy it thickens further to implant the embryo.

Q.21 [Short] What is the placenta?

During pregnancy, the vascular spaces of the endometrium fuse and become interconnected, forming the placenta. It supplies oxygen and nutrition to the embryo and foetus.

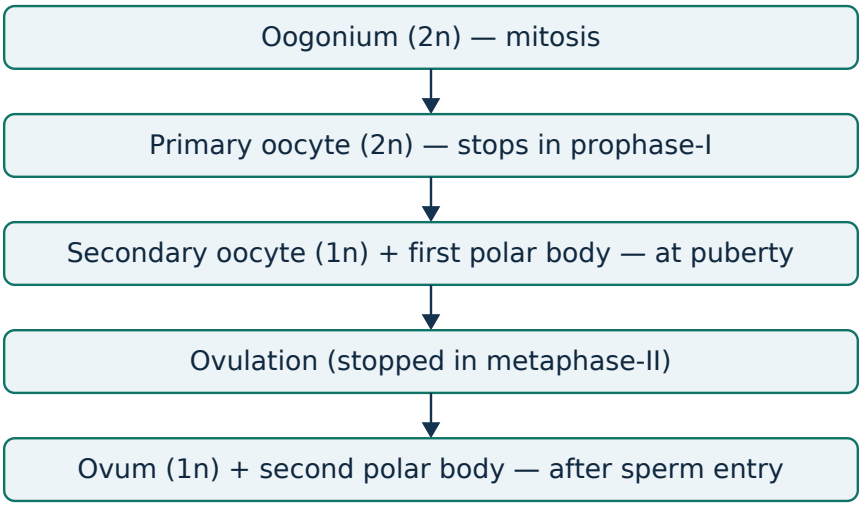
Q.22 [Short] Write a note on the vagina.

It is a thin-walled, 8–10 cm long tube leading from the outside to the cervix of the uterus. It provides a passageway for the delivery of the infant and for menstrual flow.

Q.23 [Long] What is oogenesis? Describe its steps.

1. The production of the ovum starts during the embryonic development of the female. In that period, the **oogonia** (2n) present in the follicles divide by mitosis to produce **primary oocytes** (2n).
2. The primary oocytes begin the first meiotic division, but stop in prophase-I. They remain in prophase-I all through childhood (for at least 10 to 14 years).
3. At puberty, only one of the many primary oocytes in the ovary resumes meiosis-I and produces two dissimilar haploid cells. The larger cell is the **secondary oocyte** (1n) and the smaller one is the **first polar body** (1n).
4. The secondary oocyte stops division in metaphase-II. It is released from the ovary (ovulation).
5. If the ovulated secondary oocyte is not fertilized by a sperm, it dies. If a sperm fuses with it, it quickly completes meiosis-II (in the oviduct) and produces one large **ovum** (1n) and a small **second polar body** (1n).

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Oogenesis

The textbook sentence about the secondary oocyte reads "It released from the body"; it is released from the ovary.

Q.24 [Short] How many follicles are present in the ovaries at birth?

At birth, a female's ovaries contain some 2 million follicles, each with a primary oocyte.

Q.25 [Short] Differentiate between spermatogenesis and oogenesis.

Spermatogenesis	Oogenesis
Production of sperms in the seminiferous tubules.	Production of the ovum in the ovarian follicles.
One spermatogonium produces four spermatids.	One primary oocyte produces one ovum and polar bodies.
Meiosis proceeds without stopping.	Meiosis stops in prophase-I and in metaphase-II.
Meiosis-II is completed in the testis.	Meiosis-II is completed in the oviduct, only after a sperm fuses.

17.1.3 FEMALE REPRODUCTIVE CYCLE – THE MENSTRUAL CYCLE

Q.26 [Long] Describe the events of the menstrual cycle and its hormonal regulation.

The female reproductive system functions by going through cyclic events, collectively called the **menstrual cycle**. It lasts approximately one month (28 days on average). It has the following phases.

1. Menstrual phase (Days 1–5)

The functional layer of the endometrium is shed. The detached tissue and blood move out of the body. This is called **menstruation**. It continues for 3–5 days. It occurs due to low levels of estrogen and progesterone. FSH begins to rise in this phase.

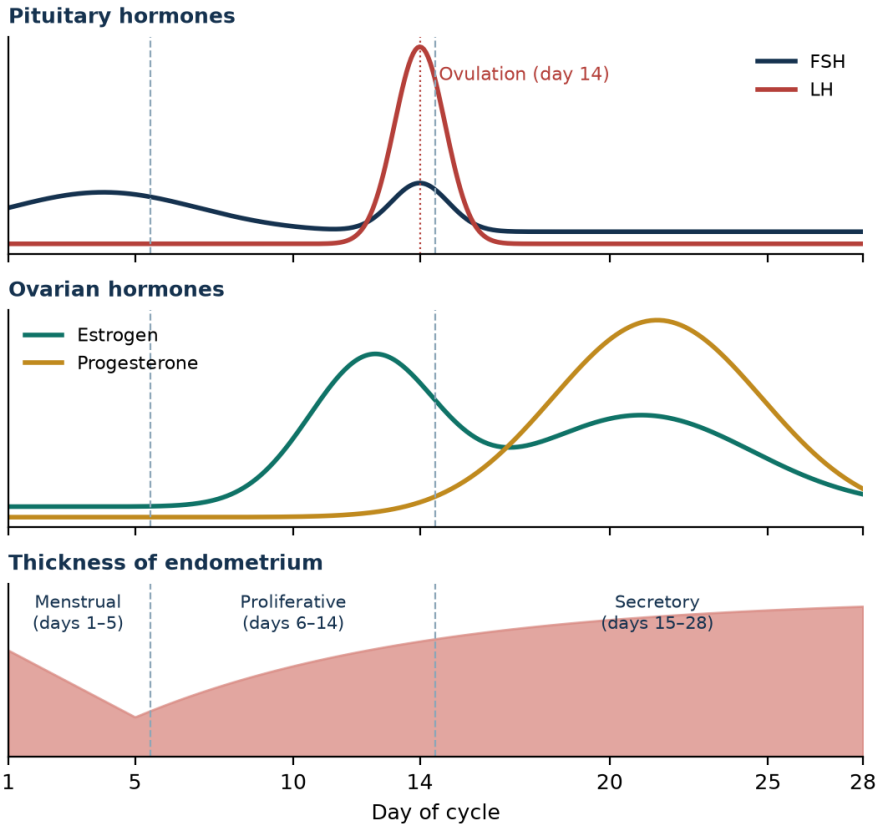
2. Proliferative phase and ovulation (Days 6–14)

Due to the rise in FSH, a few ovarian follicles begin growing, but only one grows fully and the others degenerate. This mature follicle is called the **Graafian follicle**. FSH stimulates it to secrete **estrogen**, which causes the initial vascularization of the endometrium and makes it velvety and thick again. As the concentration of estrogen rises, the level of FSH falls due to negative feedback, and the anterior pituitary releases **LH** at the end of this phase (day 14). The sudden release of LH stimulates the Graafian follicle to release the ovum into the oviduct. This is **ovulation**.

3. Secretory / post-ovulatory phase (Days 15–28)

After ovulation, LH stimulates the ruptured Graafian follicle to develop into a yellow structure called the **corpus luteum**. It secretes estrogen and **progesterone**, which make the endometrium more vascular and glandular.

If fertilization does not occur, the corpus luteum starts to degenerate and the anterior pituitary stops secreting LH. The destruction of the corpus luteum causes an abrupt fall in estrogen and progesterone. So the thickened endometrium is shed with bleeding, i.e., menstruation on day 28.



Hormone levels and endometrium during the menstrual cycle (schematic)

Hormone	Source	Role in the menstrual cycle
FSH	Anterior pituitary	Growth of ovarian follicles; stimulates the Graafian follicle to secrete estrogen
Estrogen	Graafian follicle (later corpus luteum)	Thickening and vascularization of the endometrium; lowers FSH by negative feedback

Hormone	Source	Role in the menstrual cycle
LH	Anterior pituitary	Causes ovulation (day 14); changes the ruptured follicle into the corpus luteum
Progesterone	Corpus luteum	Makes the endometrium more vascular and glandular; its fall causes menstruation

The textbook writes "The gonadotropins and FSH begin to rise in this phase" and "Due to the fall in FSH level, the anterior pituitary releases LH".

Q.27 [Short] What is menstruation? Why does it occur?

Menstruation is the shedding of the functional layer of the endometrium; the detached tissue and blood move out of the body for 3–5 days. It occurs due to low levels of estrogen and progesterone.

Q.28 [Short] What is the Graafian follicle?

Due to the rise in FSH, a few ovarian follicles begin growing, but only one grows fully. This mature follicle is called the Graafian follicle. It rises to the surface of the ovary, secretes estrogen and releases the ovum at ovulation.

Q.29 [Short] What is the role of the corpus luteum?

It is the yellow structure formed from the ruptured Graafian follicle under the influence of LH. It secretes estrogen and progesterone, which make the endometrium more vascular and glandular. If fertilization does not occur, it degenerates and menstruation follows.

Q.30 [Short] What is hCG? Why is it tested in pregnancy tests?

When the ovulated oocyte is fertilized, the tiny embryo secretes human chorionic gonadotropin (hCG). It maintains the corpus luteum, so estrogen and progesterone remain high and menstruation is prevented. Because hCG comes from the embryonic chorion and not from the mother, its presence shows pregnancy.

Q.31 [Short] What is menopause?

In human females, the menstrual cycle stops around 50 years of age. This is termed menopause.

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